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South East Technological University

AI PRINCIPLES AND PRACTICE · WEEK 9

Adapt, Do Not Rebuild

Core booklet: transfer learning, foundation models and data augmentation

Plain English first. Every formula written three ways: in symbols, in words, and with numbers you can check on your phone.

Lecturer: Indrakshi Dey

How to use this booklet

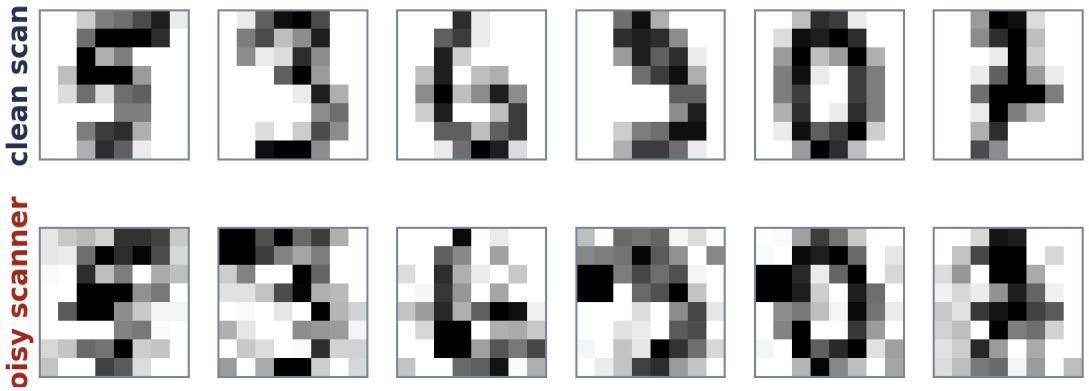
This booklet goes with the Week 9 lecture and lab. It follows the lecture's four parts in the same order, with the same example, and slows each one down: every picture is explained, every formula is written three ways, and every number comes from the lab, so you can check it yourself.

REUSE	ADAPT	AUGMENT	COUNT THE COST
don't start from zero	*change a little*	*data, not labels*	*not just accuracy*
pretrained models	**probe · fine-tune · LoRA**	**invariances · synthetic data**	**compute · energy · carbon**

> **KEY IDEA**
> Don't start from zero: reuse a model that already knows a lot, change it a little, and teach it with data, not labels. And count what it costs.

The example all week. A small network was trained on 900 clean scanned digits; it stands in for a big pretrained model. The new job: digits from a cheap, noisy scanner, with only 5 labelled examples of each digit.

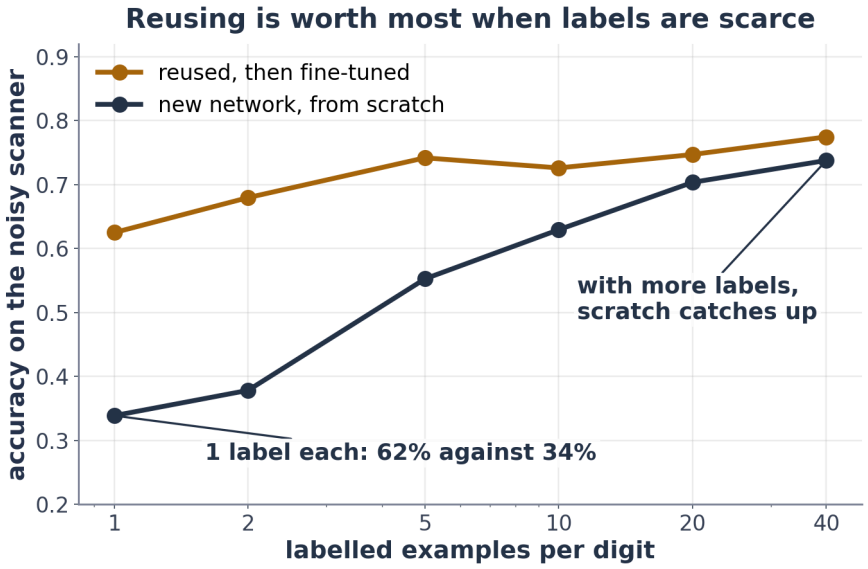
The same digits, from a cheaper scanner



Each section has the same boxes. **In plain English** says the idea with no symbols. **Everyday analogy** ties it to something familiar. The formula box writes it three ways. **Worked example** does the arithmetic. **Keep this** marks what to remember.

Part 1 Reuse: don’t start from zero

1.1 Reusing beats starting again, when labels are few



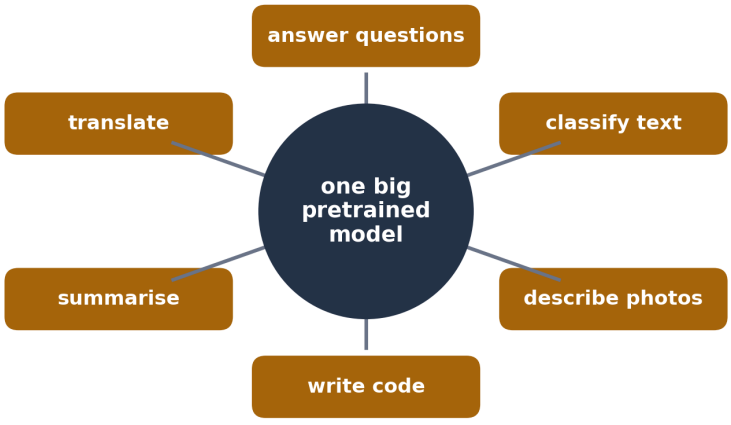
IN PLAIN ENGLISH

With 5 labels per digit, a brand-new network trained from scratch gets 55% of the noisy scanner’s digits right. The reused network, with only its last layer retrained, gets 71%. With 1 label per digit the gap is bigger still: 62% against 34%. With 40 labels per digit, scratch has nearly caught up. Reusing is worth most when labels are scarce, which in real projects is almost always.

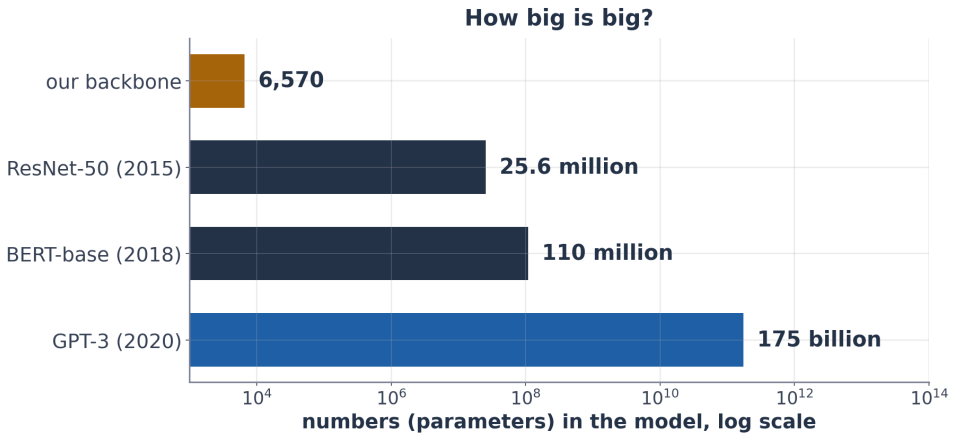
EVERYDAY ANALOGY

If you can drive a car, you do not take driving lessons from the beginning to drive a van. You reuse almost everything and learn the few things that differ.

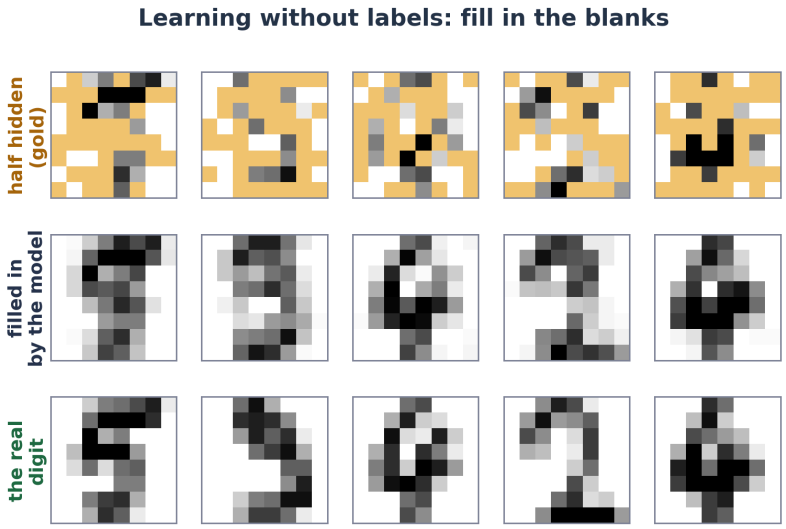
1.2 Foundation models



A **foundation model** is one big model trained once, on a huge amount of data, at huge cost, and then reused for many jobs. Training from scratch for every new job is now the exception.



1.3 How they learn without labels: fill in the blanks



Nobody labels the whole internet. Foundation models learn by hiding part of each example and predicting it from the rest: the next word, a masked word, or a hidden patch of an image. The labels come free from the data itself. This is **self-supervised learning**. Above, a network was shown digits with half their pixels hidden and learned to fill them in, without ever seeing a label.

THE FORMULA $$L = -\,\text{average of}\,\log p(\text{hidden} \mid \text{visible})$$

IN WORDS **loss = the average surprise at each hidden piece, given the pieces you can see**

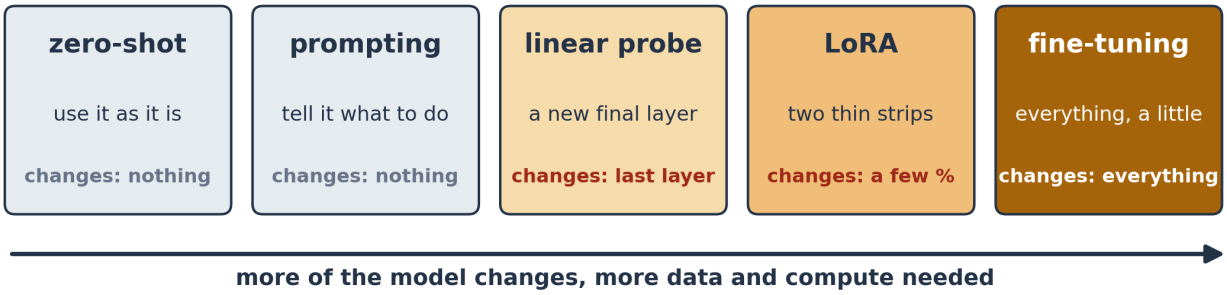
WITH NUMBERS **“the cat sat on the …”: “mat” given 40%, so the surprise is $-\log_2 0.4 = 1.3$ bits**

★ KEEP THIS

It is Week 8′s cross-entropy again: the model learns by becoming less surprised by what was hidden.

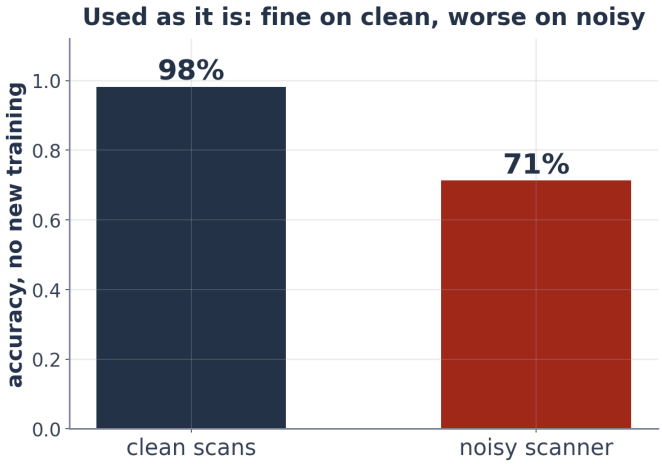
Part 2 Adapt: change a little

2.1 The transfer spectrum



From left to right, more of the model changes, and you need more labels, more memory and more compute.

2.2 Use it as it is (zero-shot)



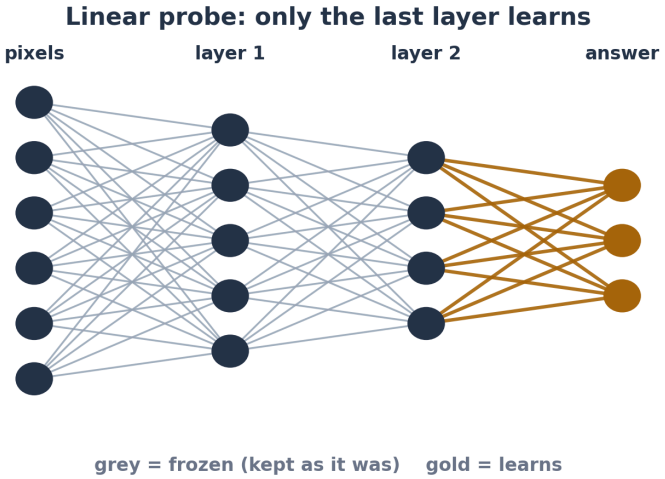
IN PLAIN ENGLISH

No new training at all. The backbone gets 98% of clean digits right but only 71% from the noisy scanner: the job is the same, but the conditions changed.

2.3 Prompting

With a language model you can change the **input**, not the model: describe the job, and perhaps show a few examples, in the prompt. No number in the model changes. Week 10 explains why this works.

2.4 Linear probe: a new head on a frozen body

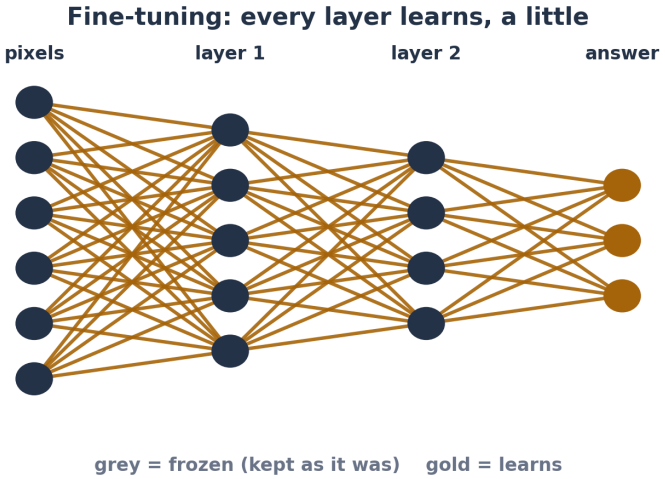


THE FORMULA $$\hat{y} = \text{softmax}(W_{\text{head}} \cdot f(x))$$

IN WORDS **answer = a new last layer applied to what the frozen network sees**

WITH NUMBERS **330 numbers trained: 71% on the noisy scanner**

2.5 Fine-tuning: every layer learns, a little

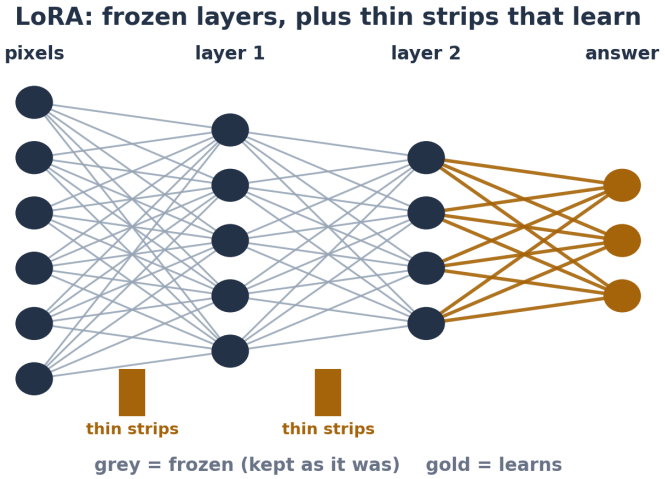


THE FORMULA $$\theta \leftarrow \theta - \eta \nabla L \text{ from } \theta_{\text{pretrained}}$$

IN WORDS **the same downhill step as Week 7, starting from the pretrained dials, with a small step**

WITH NUMBERS **all 6,570 numbers trained: 74%**

2.6 LoRA: frozen layers, plus thin strips



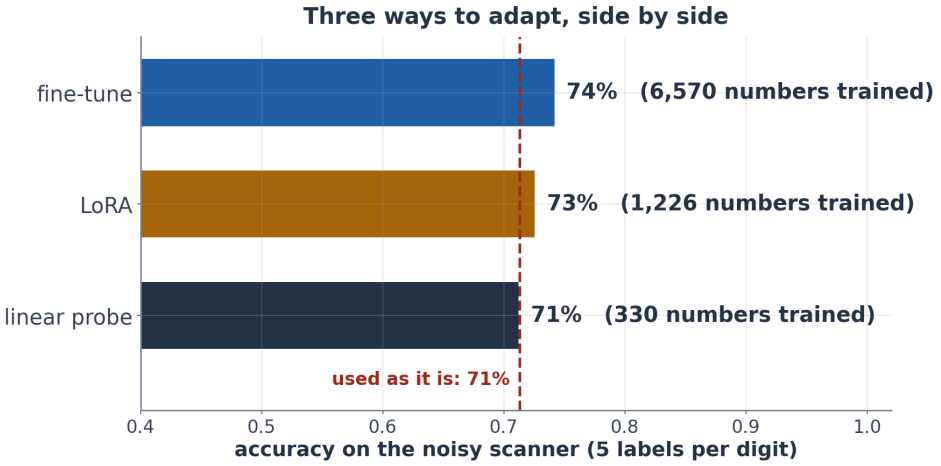
THE FORMULA $$W' = W + BA$$

IN WORDS **new weights = old weights (frozen) + thin strip × thin strip**

WITH NUMBERS **rank 4: 1,226 numbers trained, 73%**

Worked example: counting LoRA's numbers
For one 64 × 64 table at rank 4: B is 64 × 4 = 256 numbers and A is 4 × 64 = 256, so 512. For the 64 × 32 table: 64 × 4 + 4 × 32 = 384. Add the final layer's 330: 512 + 384 + 330 = **1,226**. For a 768 × 768 table at rank 8: 768 × 8 + 8 × 768 = 12,288, instead of 589,824.

2.7 Three ways, side by side



★ KEEP THIS
Change only as much as the job needs. A probe is the quick, honest baseline; LoRA is the usual choice for big models; full fine-tuning needs the most labels, memory and compute.

Part 3 Augment: data, not labels

3.1 More examples, without more labels



EVERYDAY ANALOGY

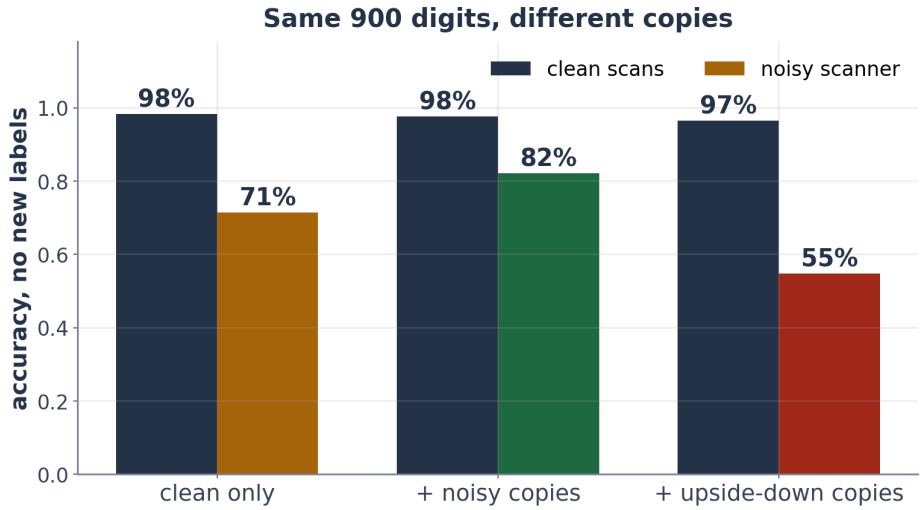
A smudged 6 is still a 6. By showing the model smudged copies with the same label, you tell it: **ignore smudges**. That is called an **invariance**.

THE FORMULA $$f(T(x)) = f(x)$$

IN WORDS **the answer should not change when you apply the change T**

WITH NUMBERS **T = add scanner noise: 71% → 82%, with no new labels**

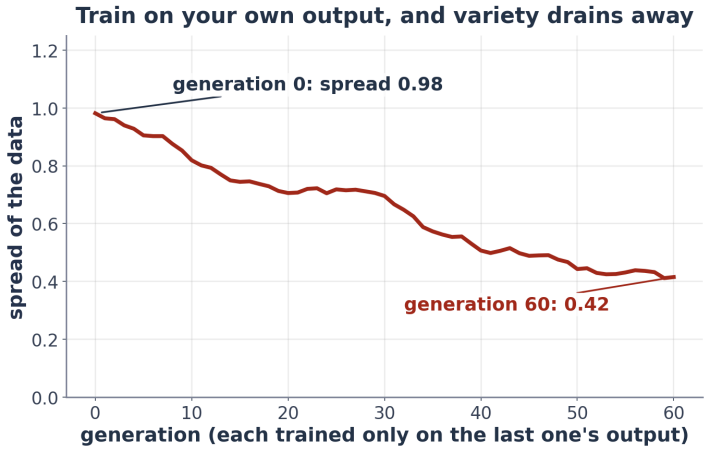
3.2 The right copies help; the wrong ones hurt



IN PLAIN ENGLISH

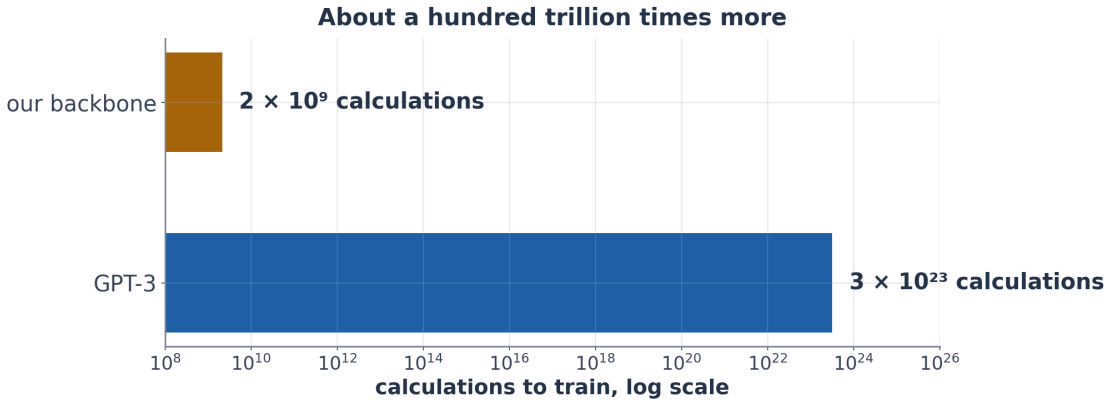
Training the same network on the same 900 clean digits plus noisy copies raised its accuracy on the noisy scanner from 71% to 82%, with no new labels. Adding upside-down copies dropped it to 55%: an upside-down 6 is a 9, so those copies teach a lie.

3.3 Synthetic data: useful, with care



Data made by a model (synthetic data) can fill gaps, protect privacy, or cover rare cases. But a model trained only on its own output slowly loses variety: above, each generation is fitted only to the previous one’s samples, and the spread shrinks from 0.98 to 0.42 in 60 generations. Check synthetic data against real data, and keep real data in the mix.

Part 4 Count the cost: not just accuracy



THE FORMULA $C \approx 6ND$

IN WORDS **calculations to train ≈ 6 × numbers in the model × examples**

WITH NUMBERS **GPT-3: 6 × 175 billion × 300 billion ≈ 3 × 10²³**

THE FORMULA $E = Pt, \quad \mathrm{CO}_2 = E \times I$

IN WORDS **energy = power × time; carbon = energy × the grid's carbon per kWh**

WITH NUMBERS **300 W for 10 h = 3 kWh ≈ 0.9 kg CO_2 (at 0.3 kg per kWh)**

Worked example: GPT-3's training, as published

Patterson and others (2021) estimated about 1,287 MWh of electricity and 552 tonnes of CO_2 for training GPT-3 once. That is 1,287,000 kWh: at 0.3 kg per kWh the same energy would give about 386 tonnes, so the published figure also reflects the particular data centre and grid. LoRA, by contrast, cut the numbers to train for GPT-3 by about 10,000 times (Hu and others, 2021).

★ KEEP THIS

Choose a model by accuracy **and** by numbers trained, memory, time, energy and carbon. On big models the cost gap is huge; the accuracy gap often is not.

Practice questions

1. When does reusing a pretrained model help most?

When labels are scarce: at 1 label per digit, 62% against 34% from scratch.

2. What is self-supervised learning?

Learning by predicting hidden parts of the data from the rest, so the labels come free from the data.

3. Order the transfer spectrum from least to most change.

Zero-shot, prompting, linear probe, LoRA, fine-tuning.

4. In a linear probe, which numbers learn?

Only the new final layer; the rest is frozen.

5. LoRA on a 1,000 × 500 table at rank 4: how many numbers?

1,000 × 4 + 4 × 500 = 6,000, instead of 500,000.

6. What does augmentation teach a model?

What does not change the answer (an invariance).

7. Why do upside-down copies hurt a digit model?

Because an upside-down 6 is a 9: they teach a false invariance.

8. What is a risk of training only on synthetic data?

The model loses variety over generations (model collapse), and errors compound.

9. Roughly how many calculations to train a 1-billion-number model on 20 billion words?

$6 \times 10^{9} \times 2 \times 10^{10} = 1.2 \times 10^{20}$.

10. A 250 W card for 4 hours, at 0.3 kg CO_2 per kWh?

1 kWh, so 0.3 kg CO_2.

11. Name two reasons to prefer LoRA over full fine-tuning of a large model.

Far less memory and compute, and a tiny file per job instead of a full copy of the model.

Glossary

augmentation	Adding changed copies of examples, with the same labels, to teach what does not matter.
backbone	The pretrained part of a model that is reused.
fine-tuning	Training all of a pretrained model’s numbers a little on the new job.
foundation model	One big model trained once on huge data and reused for many jobs.
invariance	A change that should not change the answer: f(T(x)) = f(x).
linear probe	Training only a new final layer on a frozen pretrained model.
LoRA	Low-rank adaptation: learning thin strips B and A added to frozen weights, W + BA.
model collapse	Loss of variety when models are trained on their own outputs.
prompting	Changing the input (instructions, examples) instead of the model.
self-supervised learning	Learning by predicting hidden parts of the data; no human labels.
synthetic data	Data generated by a model or a simulator rather than collected.
transfer learning	Reusing what a model learned on one job for another.
zero-shot	Using a pretrained model as it is, with no new training.